

Course Name:	Basic of Electrical Engineering Laboratory (BEEL) 316U06L102	Semester:	I
Date of Performance:	/ / 20__	Batch No:	
Faculty Name:		Roll No:	
Faculty Sign & Date:		Grade/Marks:	

Experiment No: 5

Title: Battery Voltage level Indicator.

Aim and Objective of the Experiment:

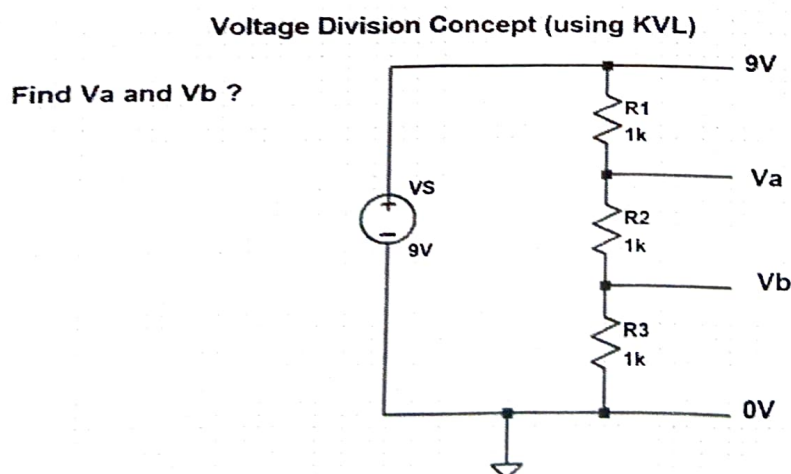
- To understand voltage division concept, current division concept and principle of operation of LED.
- To develop a micro project (Battery level indicator) based on the concepts learned in the form of various task performed in the experiment.

COs to be achieved:

CO1: Analyze resistive networks excited by DC sources using various network theorems.

Circuit Diagram/ Block Diagram:

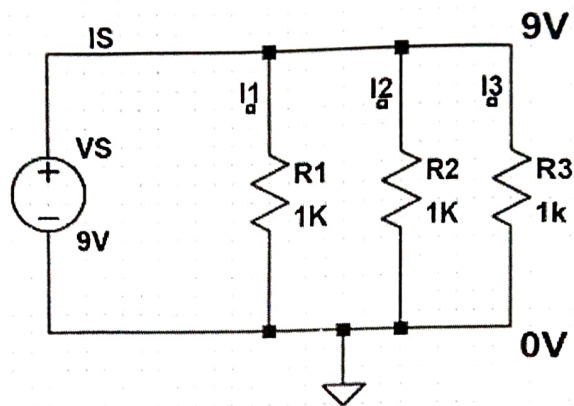
Task 1: Voltage division Concept and its verification on breadboard



Task 2: Current division Concept

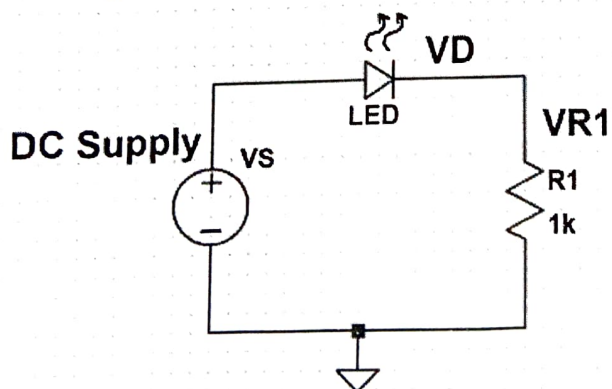
Current division concept (using KCL)

Find I_S , I_1 , I_2 and I_3 ?



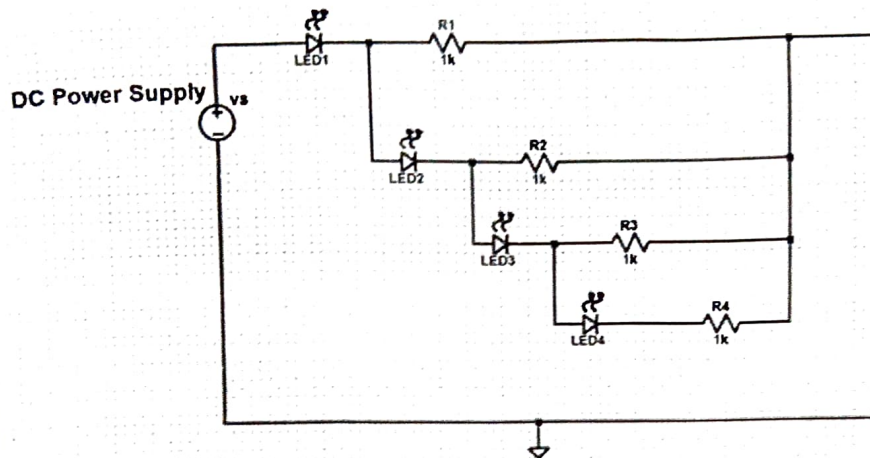
3: Turn on an LED and measure its turn-on voltage

Turning on an LED and concept of current limiting resistor



Task 4: Battery Level Indicator Circuit

Battery Level Indicator Circuit diagram



Stepwise-Procedure:

1. Make the connections as shown in the circuit diagram for Task1. Measure the voltages V_a , V_b and current I_s for Task 1 and compare with calculated results.
2. Make the connections as shown in the circuit diagram for Task2. Measure the currents I_1, I_2, I_3 and I_S and compare with calculated results.
3. Make the connections as shown in the circuit diagram for Task3. Measure the voltages V_S, V_D, V_{R1} for Case1 and Case 2.
4. Make the connections as shown in the circuit diagram for Task4. Measure the voltages across LED and resistors.

Observation Table:

Observation Table 1 (Task 1)

Voltages / Currents	Theoretical reading	Practical reading
V_a	6V	5.92V
V_b	3V	2.95V
I_s	3mA	3mA

Calculations (Task1):

Calculate V_a and V_b using the formula given below:

$$V_a = \frac{(R_2 + R_3) * V_S}{R_1 + R_2 + R_3}$$

$$V_b = \frac{(R_3) * V_S}{R_1 + R_2 + R_3}$$

Observation Table 2

Currents	Theoretical reading (I mA)	Practical Reading (I mA)
I1	9mA	9mA
I2	9mA	9mA
I3	9mA	9mA
IS	27mA	27mA

Calculations (Task2):

Calculate I1, I2, I3 and IS using the formula given below:

$$I1 = \frac{V_S}{R_1}$$

$$I2 = \frac{V_S}{R_2}$$

$$I3 = \frac{V_S}{R_3}$$

$$IS = I1 + I2 + I3$$

Observation (Task 3):

Case 1 : LED just turn's ON

Parameters	Practical reading
VS	1.6V
VD	1.58V

VR1	0.02V
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Case 2 : LED turn's ON (glows brightly)

Parameters	Practical reading
VS	3.7V
VD	1.79V
VR1	2.00V

Observations (Task4):

Case 1 : Supply Voltage Levels recording

Scenario	DC Supply voltage (VS) reading
LED 1 ON	2.0V
LED 1 ON & LED 2 ON	2.0V - 4.5V (Just on) (Bright)
LED 1 ON & LED 2 ON & LED 3 ON	4.5V - 6.0V (Just on) (Bright)
ALL LEDs ON	6.0 - 8.5V

Case 2: Status of voltages in the circuit

Voltages	Practical reading (in Volts)
V _{LED 1}	8.58V
V _{LED 2}	11.25V
V _{LED 3}	11.45V
V _{LED 4}	11.45V

VR1	7.75V
VR2	7.77V
VR3	7.8V
VR4	7.77V

Screenshot of Output:

Post Lab Subjective/Objective type Questions:

1. Mention some applications of battery level indicator.

A battery indicator is a device which gives information about a battery. This will usually be a visual indication of the battery's state of charge. Some of its applications are battery indicators in power banks, wireless earphones, laptops and many other battery appliances.

2. Explain practical usage of Voltage-division concept?

The voltage division is used only there where the voltage is regulated by dropping a particular voltage in a circuit. It mainly used in such systems where energy efficiency does not necessarily to be considered seriously. Some practical usages are in battery level indicators and in invertors where the capacity has to be indicated.

3. Explain working of Battery Level Indicator implemented in this experiment in your own words?

We used the concept of Voltage division wherein we have connected the resistors in parallel to each other so that the LEDs connected next to it only lights up when the battery is above a particular voltage. Battery level indicator lets us know the status of battery of a device just by glowing the number of LEDs. The voltage probe helps us to know the voltage difference values at which the LEDs glow dimly and brightly. Using this we can determine the status of the battery.

Conclusion:

In this experiment we learnt the uses and applications of battery level indicator and voltage division.

Signature of faculty in-charge with Date: